K.R. MANGALAM UNIVERSITY

Database Management Lab File

(ENCS205)

Bachelor of Technology

in

Computer Science and Engineering

Submitted BY

Gaurav (2401010009)

B. Tech CSE (Core) – 4th Semester

Section: A

Submitted To:

MS. Mansi Kajal

School of Engineering & Technology
K. R. MANGALAM UNIVERSITY

Sohna, Haryana 122103, India FEB 2026

LAB SHEET – 2

Objective

To understand and implement SQL transactions using COMMIT, SAVEPOINT, and ROLLBACK, retrieve records using SELECT queries, and perform INSERT, UPDATE, and ALTER operations on tables.

DDL (Data Definition Language)

1. Create Table CLASS

```
CREATE TABLE CLASS (
   ID NUMBER PRIMARY KEY,
   NAME VARCHAR(20)
);
```

Initial Records:

```
INSERT INTO CLASS VALUES (1, 'Bravo');
INSERT INTO CLASS VALUES (2, 'Alex');
INSERT INTO CLASS VALUES (4, 'Cheng');
COMMIT;
```

2. Create Table EMPLOYEES

```
CREATE TABLE EMPLOYEES (
   Employee_Id NUMBER PRIMARY KEY,
   First_Name VARCHAR(20),
   Last_Name VARCHAR(25),
   Email VARCHAR(50),
   Phone_Number VARCHAR(20),
   Hire_Date DATE,
```

```
   Job_Id VARCHAR(20),
   Salary NUMBER(8,2),
   Commission_Pct NUMBER(2,2),
   Manager_Id NUMBER,
   Department_Id NUMBER
);
```

3. Create Table Client_master

```
CREATE TABLE Client_master (
   ClientNo VARCHAR(6) PRIMARY KEY,
   Name VARCHAR(20),
   City VARCHAR(20),
   Pincode NUMBER(8),
   State VARCHAR(20),
   Bal_due NUMBER(10,2)
);
```

4. Rename Client_master to Client12

```
RENAME Client_master TO Client12;
```

5. Display bal_due as BALANCE

```
SELECT Bal_due AS "BALANCE" FROM Client_master;
```

DML (Data Manipulation Language)

Part A: Transactions and Savepoints (CLASS Table)

```
INSERT INTO class VALUES (5,'Rahul');
COMMIT;

UPDATE class SET name = 'Abhijeet' WHERE id=5;
SAVEPOINT A;
```

```
INSERT INTO class VALUES (6,'Chris');
SAVEPOINT B;

INSERT INTO class VALUES (7,'Bravo');
SAVEPOINT C;

SELECT * FROM class;

ROLLBACK TO B;
SELECT * FROM class;

ROLLBACK TO A;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

ID	NAME
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
6	Chris
7	Bravo
NULL	NULL

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

ID	NAME
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
6	Chris
NULL	NULL

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

ID	NAME
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
NULL	NULL

Part B: Retrieving Records (EMPLOYEES Table)

1. SELECT Employee_Id, First_Name, Last_Name, Salary
 FROM EMPLOYEES;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Salary
100	Raj	Mehra	9000.00
101	Rahul	Sharma	5000.00
102	Aman	AUSTIN	4800.00
103	Neha	Verma	7000.00
104	Riya	Singh	4500.00
NULL	NULL	NULL	NULL

2. SELECT *
FROM EMPLOYEES
 WHERE Manager_Id = 100;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Comm
101	Rahul	Sharma	rahul@gmail.com	9876543211	2021-02-15	IT_PROG	5000.00	0.10
102	Aman	AUSTIN	aman@gmail.com	9876543212	2022-03-20	HR_REP	4800.00	0.05
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Salary >= 4800;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

First_Name	Last_Name
Raj	Mehra
Rahul	Sharma
Aman	AUSTIN
Neha	Verma

4. SELECT *
FROM EMPLOYEES
WHERE Last_Name = 'AUSTIN';

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commis
102	Aman	AUSTIN	aman@gmail.com	9876543212	2022-03-20	HR_REP	4800.00	0.05
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

5. SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Department_Id IN (60,70,80);

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

First_Name	Last_Name
Rahul	Sharma
Aman	AUSTIN
Neha	Verma
Riya	Singh

6. SELECT DISTINCT Manager_Id
FROM EMPLOYEES;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Manager_Id
NULL
100
101
102

Part C: Updating Records (Client_master Table)

1. Insert five records:

('C101','Rahul','MG Road','Delhi','Delhi',110001,6000),

('C102','Amit','Park Street','Kolkata','WB',700016,4500),

('C103','Priya','Brigade Road','Bangalore','KA',560001,8000),

('C123','Neha','FC Road','Pune','MH',411004,3000),

('C105','Arjun','Anna Salai','Chennai','TN',600002,7000);

COMMIT;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

ClientNO	Name	Address	City	State	Pincode	Bal_due
C101	Rahul	MG Road	Delhi	Delhi	110001	6000.00
C102	Amit	Park Street	Kolkata	WB	700016	4500.00
C103	Priya	Brigade Road	Bangalore	KA	560001	8000.00
C105	Arjun	Anna Salai	Chennai	TN	600002	7000.00
C123	Neha	FC Road	Pune	MH	411004	5100.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL

2. SELECT Name
FROM Client_master
WHERE Bal_due > 5000;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name
Rahul
Priya
Arjun

3. UPDATE Client_master
SET Bal_due = 5100
 WHERE ClientNo = 'C123';

COMMIT;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

BALANCE
4500.00
6000.00
3000.00
7500.00
4800.00